

Tilak Patel

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WORK EXPERIENCE

Citadel

Nov. 2024 – Present

Software Engineer - Trading Fee Reconciliations

New York, NY

- Led complete rework of settlement fails process to correctly accrue settlement fails charges
 - Resulted in \$600,000 annual savings through proper identification of invalid counterparty claims
 - Automated monthly counterparty claim issuance for 150 counterparties by integrating with new remittance platform (below), resulting in 48 monthly operational hours saved
 - Replaced manual spreadsheet-based reconciliation process with fully automated workflow
- Executed development of new remittance platform for automated invoice generation and email chasers
 - Eliminated dependency on external vendor software, eliminating \$70,000 annual recurring charges
 - Allowed for automated invoice generation and email chasers, resulting in 16 monthly operational hours saved
- Revamped fees platform for trading desk onboarding
 - Enabled firm-wide fee charges to be properly allocated to trading desks
 - Eliminated manual ticket-based process, resulting in 12 monthly hours saved
 - Allowed full visibility into trading fees for desks, enabling data-driven negotiations with counterparties
- Integrated external vendor software into fees platform to replace manual processes
 - Automated margin interest settlement into fees platform to replace manual upload/download process, saving 15 monthly operational hours
 - Integrated procurement vendor system with fees platform to eliminate duplicate payment risk, eliminating duplicate payment errors exceeding \$50,000 annually

Twilio

Jun. 2021 – Nov. 2024

Senior Software Engineer - Billing Team

San Francisco, CA

- Led API implementation for a customer-facing Java microservice that serves the most traffic across the entire billing organization
 - Correctness of implementation enabled accurate reporting of Twilio's financial data to financial regulators
 - Led service and traffic-splitting strategies to onboard all Twilio customers to the new service silently and without disruption
- Directed containerization efforts of legacy software systems to enable migration to Kubernetes
 - Established a paved-path migration scheme to move older applications on to Docker which was utilized by the entire company
 - Worked under tight deadlines to complete the migration due to Amazon dropping support for older versions of Amazon Linux
- Architected zero-downtime cross-account migration of EC2 database system to AWS managed solution

EDUCATION

University of Maryland

May, 2021

B.S. Computer Science

College Park, MD

- GPA: 3.85 / 4

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** AWS Certified Developer - Associate, Oracle Certified Java SE8 Associate
- **Skills:** Java, Python, Amazon Web Services (AWS), Google Cloud, Linux, Spring, SQL, Docker, Kubernetes, Terraform, Kafka, Redis, Protocol Buffers, Data Structures & Algorithms
- **Interests:** Basketball, Football, Chess, Meditation, Cooking